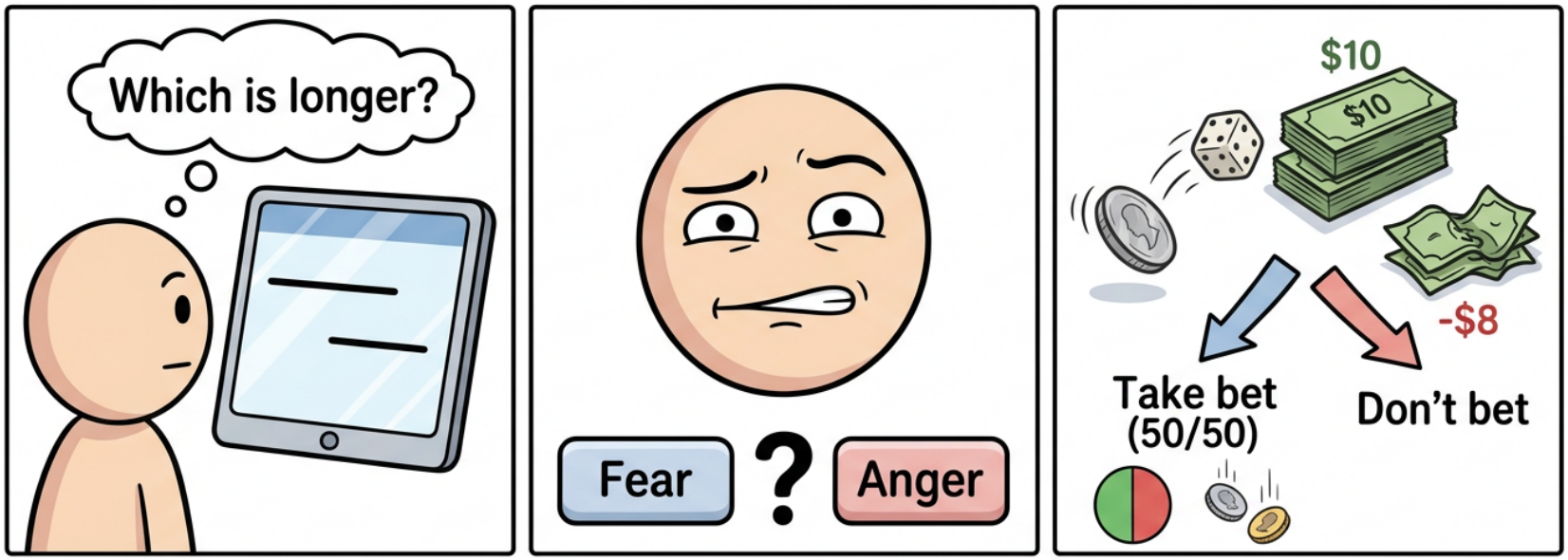
The Logistic Choice Model

Michael D. Lee

Two-alternative Forced-choice Tasks

- Two-alternative forced-choice tasks ("2AFC" tasks) involve a series of trials requiring a choice between two alternatives
  - psychophysical tasks: "was the line on the left or the right longer?" (Pelli & Farell, 1995)
  - classification tasks: "is this face showing the emotion of fear or anger?" (Wang et al., 2017)
  - gambling tasks: "would you rather have a 50% chance of winning \$10 along with a 50% chance of losing \$8, or would you rather not take that bet?" (Kahneman & Tversky, 1984)

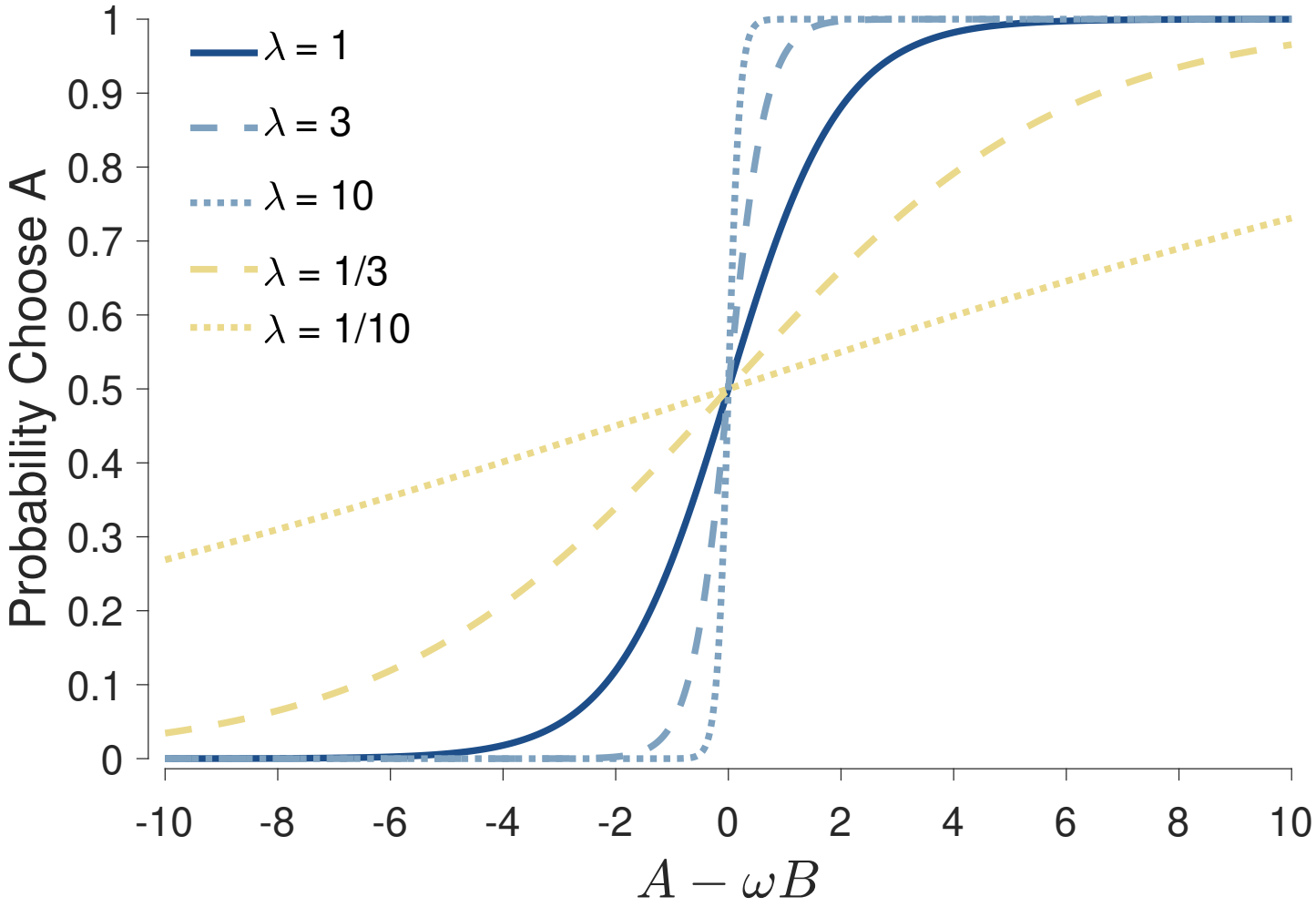


The Logistic Choice Model

- The logistic choice model assumes there is evidence A for alternative A and B for alternative B
- Instead of always choosing the alternative with more evidence, the model incorporates
  - weight: a parameter ω gives different weights to the evidence total, so that the decision is based on $A - \omega B$
  - consistency: a parameter λ controls whether the favored alternative is always chosen, or whether choice is stochastic

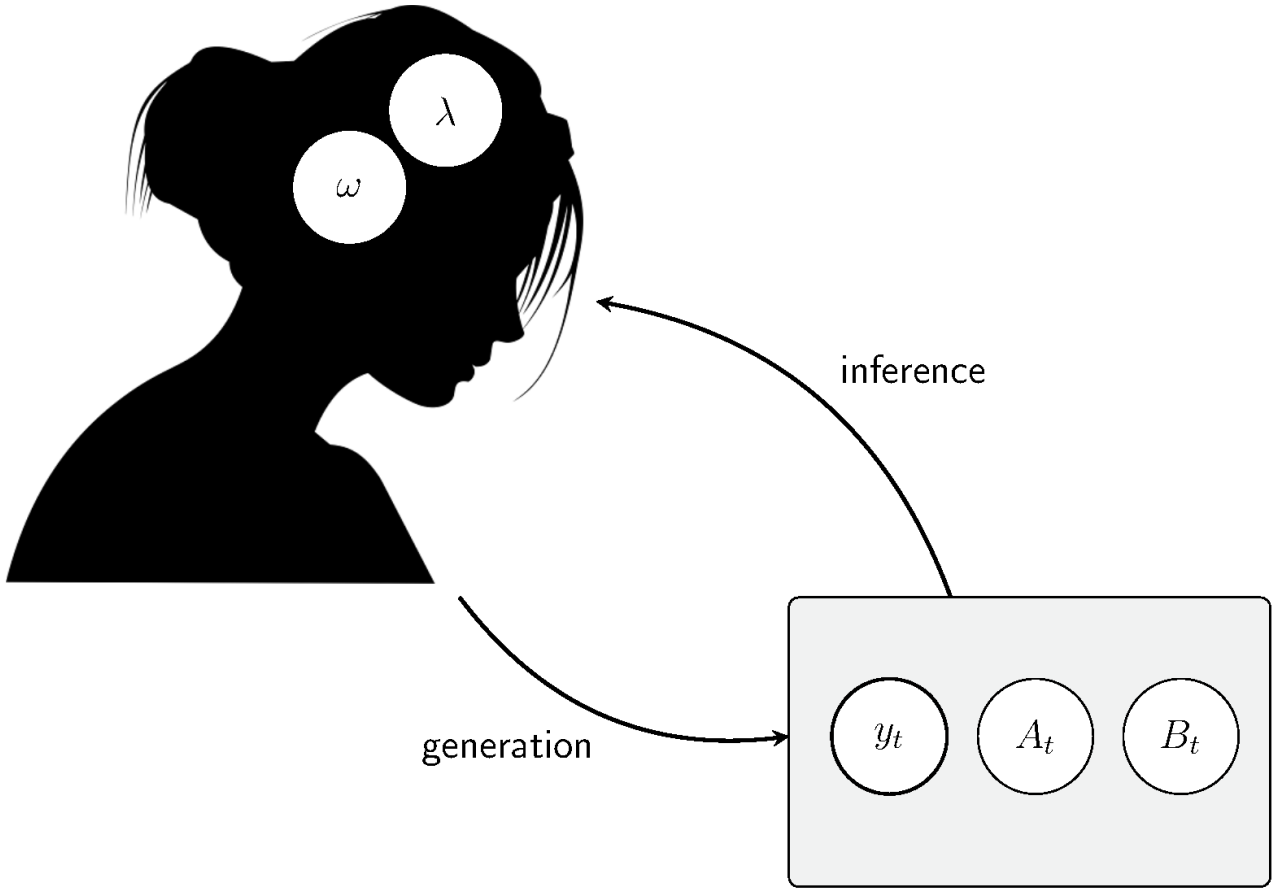
The Logistic Choice Model

- The probability of choosing alternative A is a logistic function of the weighted difference $A - \omega B$



Logistic Choice Model Structure

- The logistic choice model has the structure
  - observed binary decisions y_t on trial t for alternatives with evidence A_t and B_t
  - weight ω and consistency λ parameters



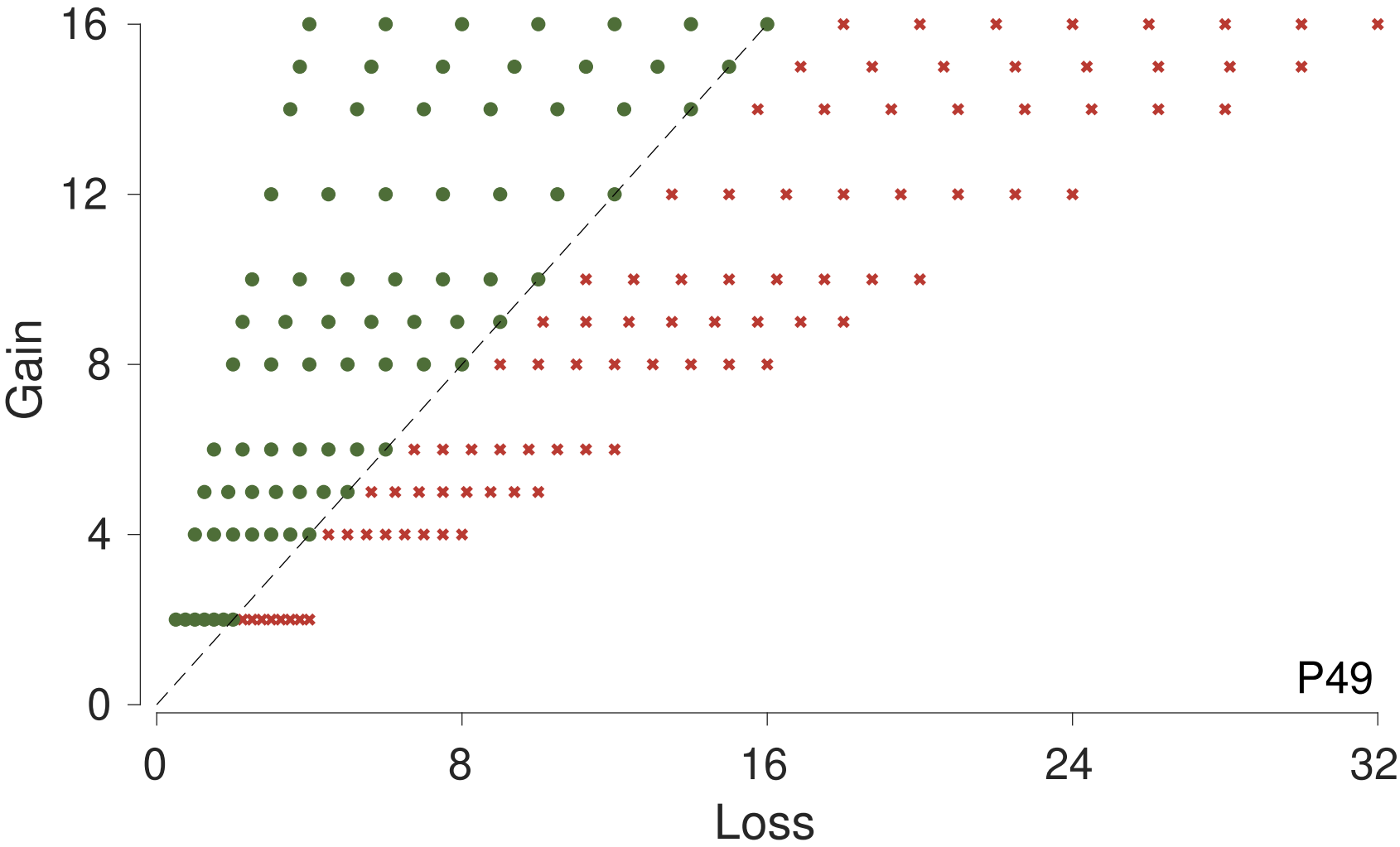
Mixed Gambles

Gambling Experiment

- Stillman (2021) provides data from a gambling experiment involving 148 participants each doing 165 trials
- On each trial there were two options
  - accept a 50-50 gamble between a gain of $\$G$ and a loss of $\$L$
  - reject the gamble
- The gains ranged from \$2 to \$16 and the losses ranged from \$0.50 to \$32

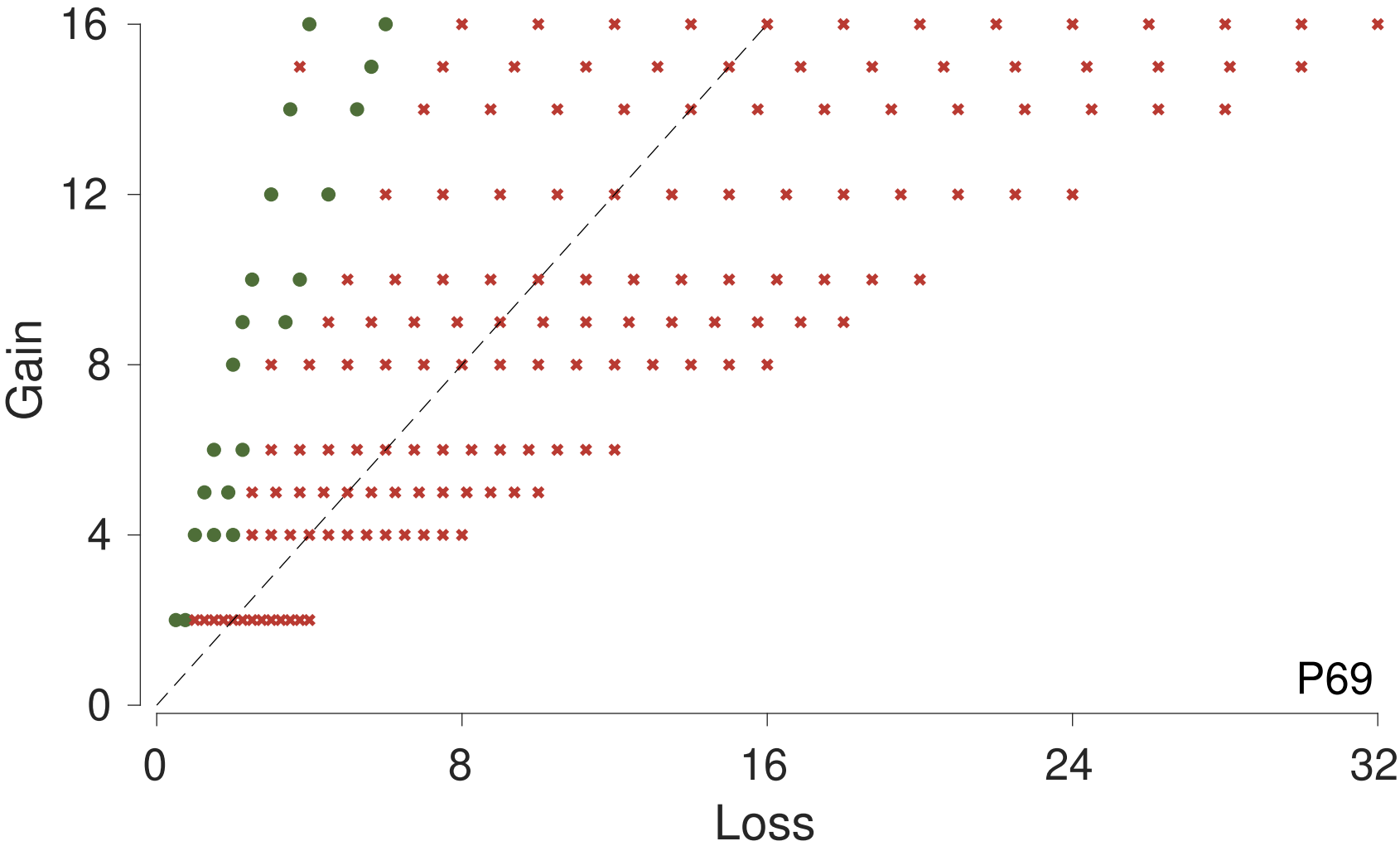
Participant B Behavior

- Each dot is a (G, L) gamble
  - green circles show gambles the participant accepted and the red crosses show those they rejected



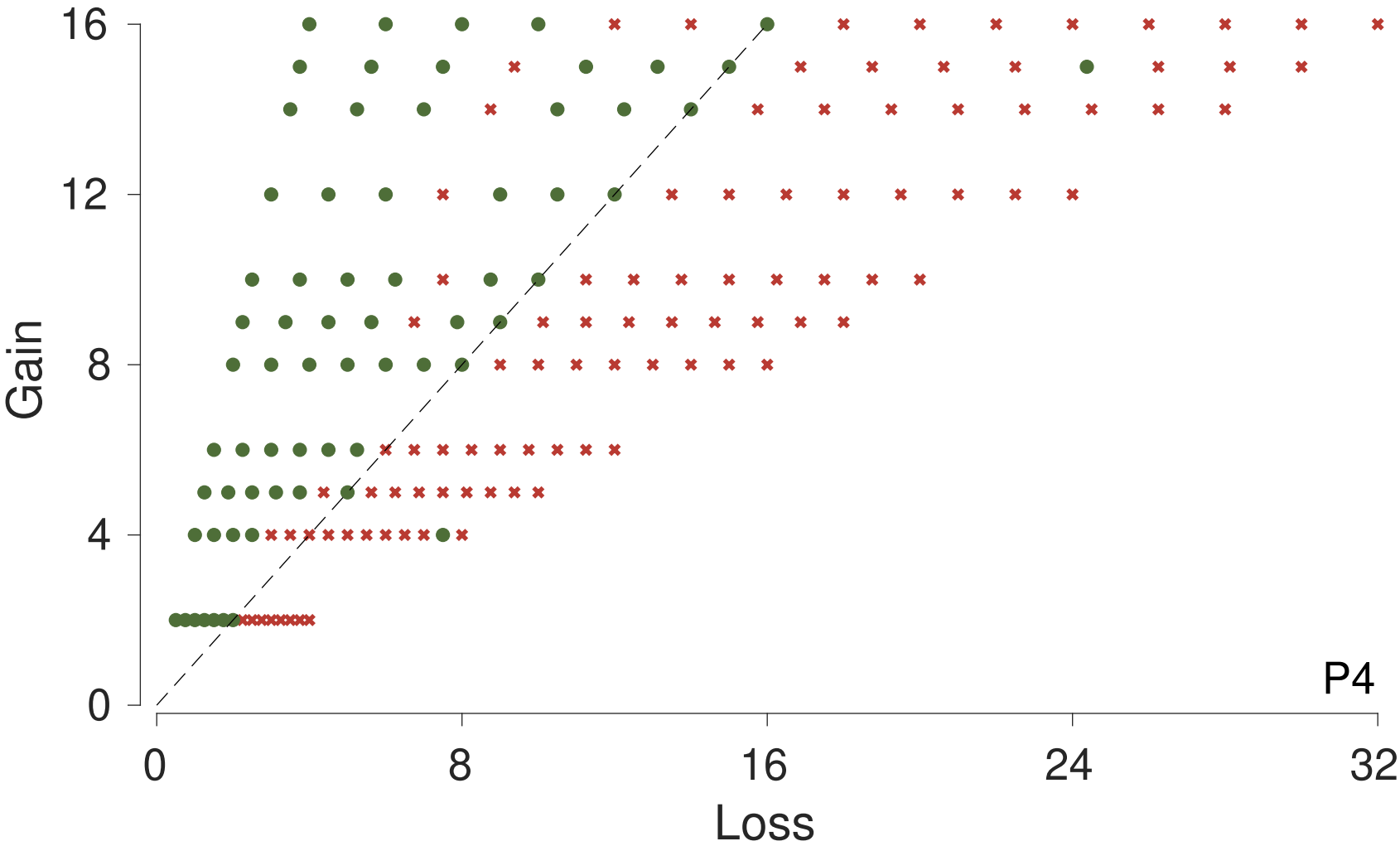
Participant J Behavior

- Each dot is a (G, L) gamble
  - green circles show gambles the participant accepted and the red crosses show those they rejected

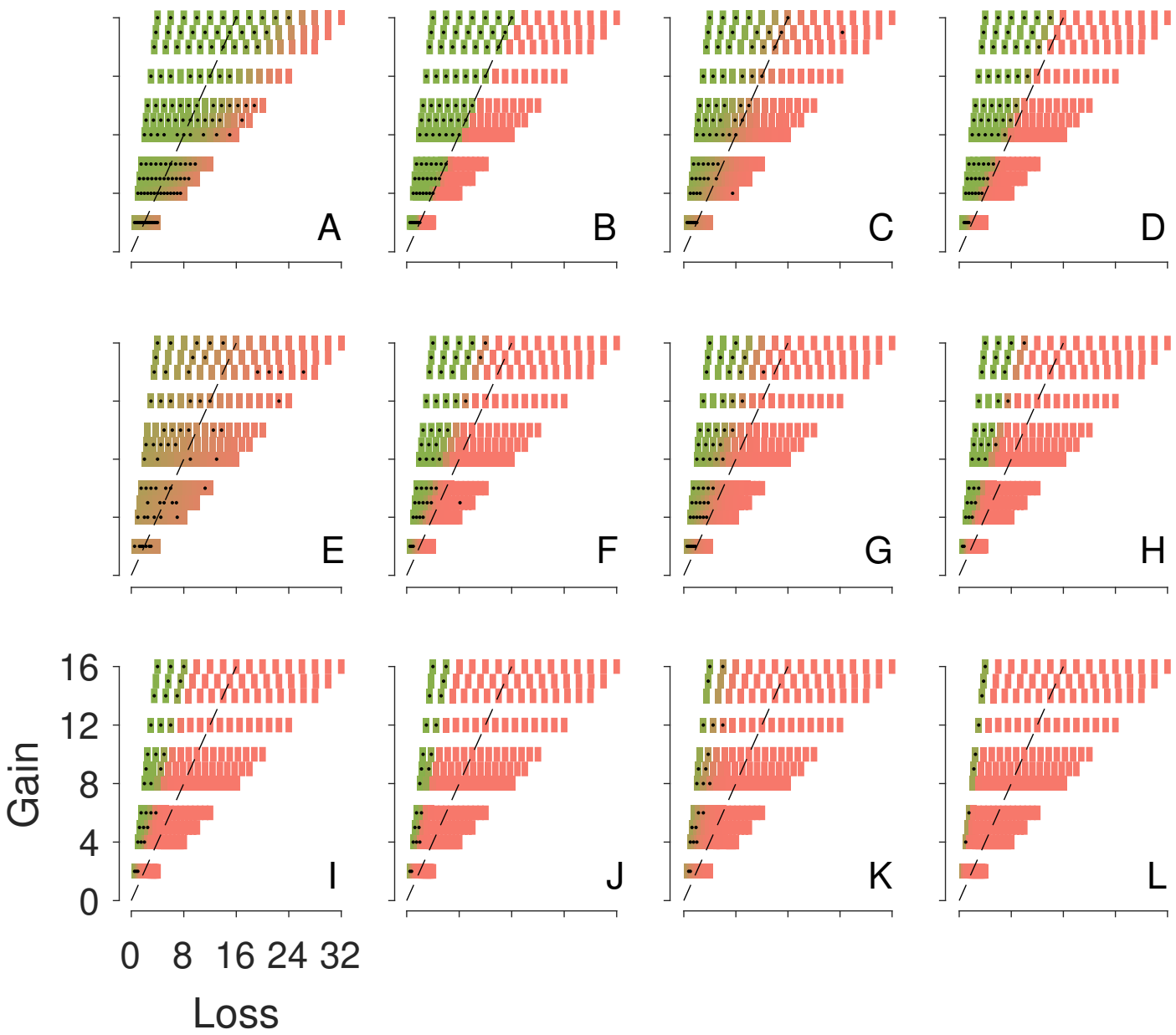


Participant C Behavior

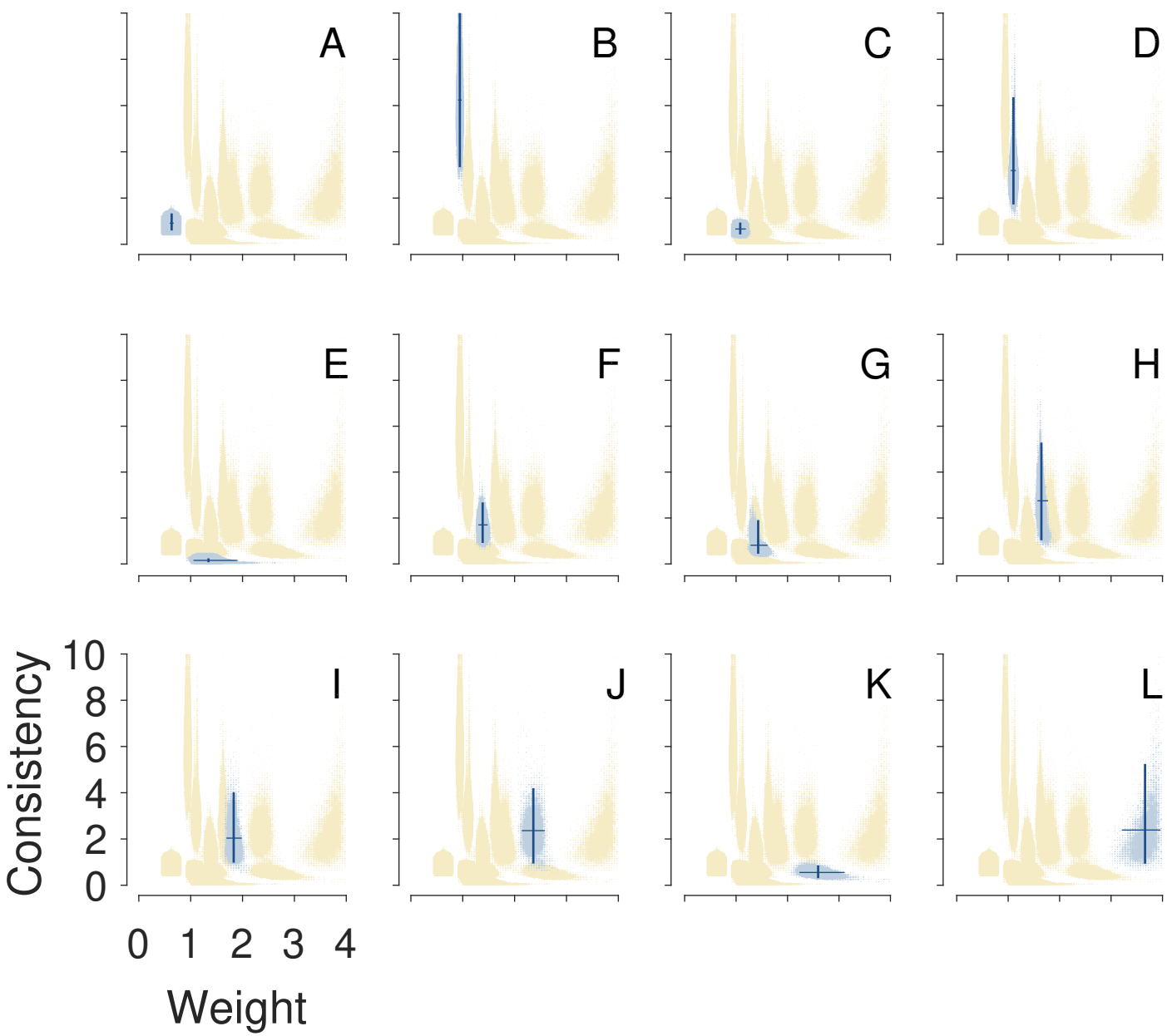
- Each dot is a (G, L) gamble
  - green circles show gambles the participant accepted and the red crosses show those they rejected



Descriptive Adequacy for 12 Participants

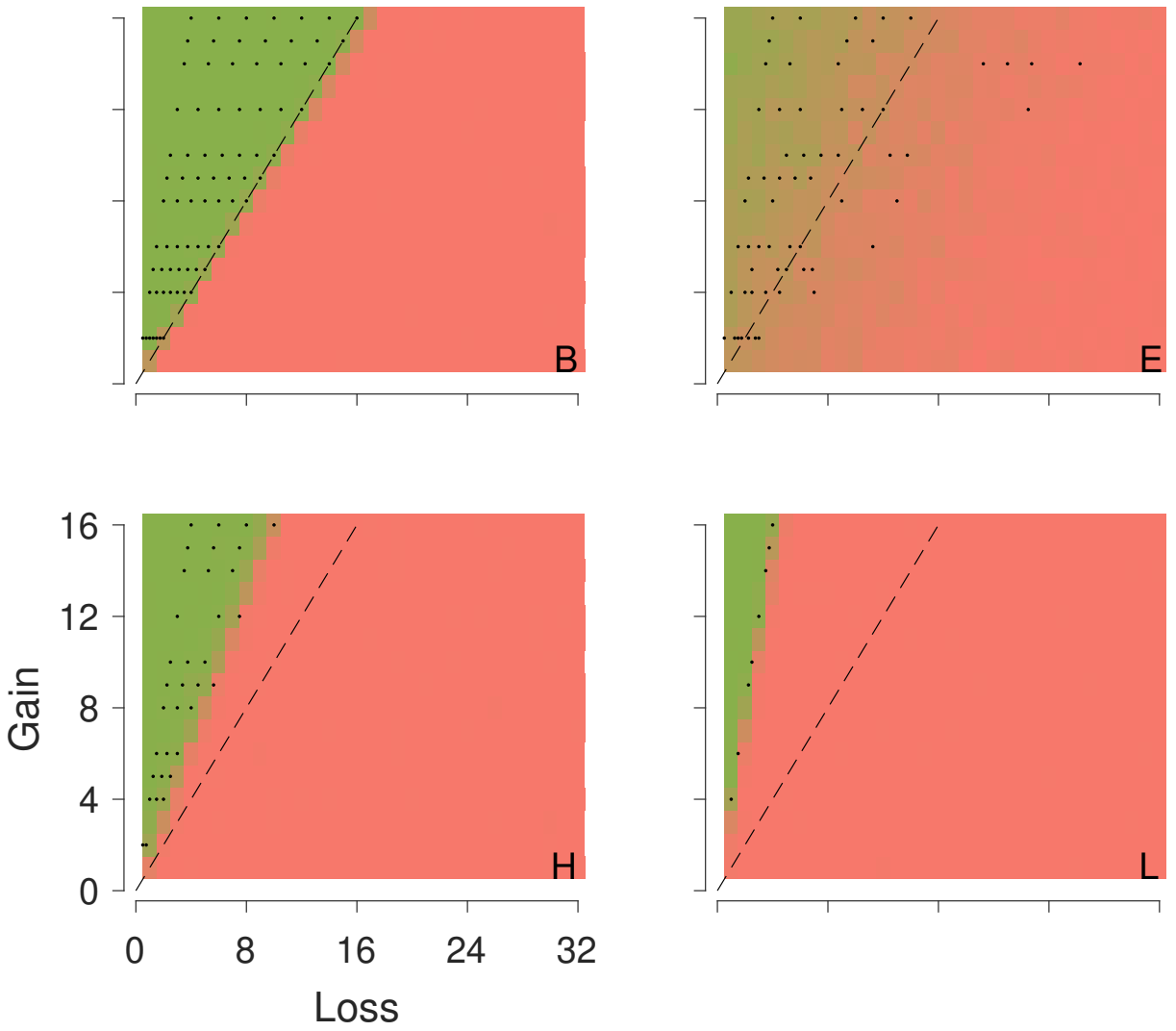


Inferences for 12 Participants



Generalization for 4 Participants

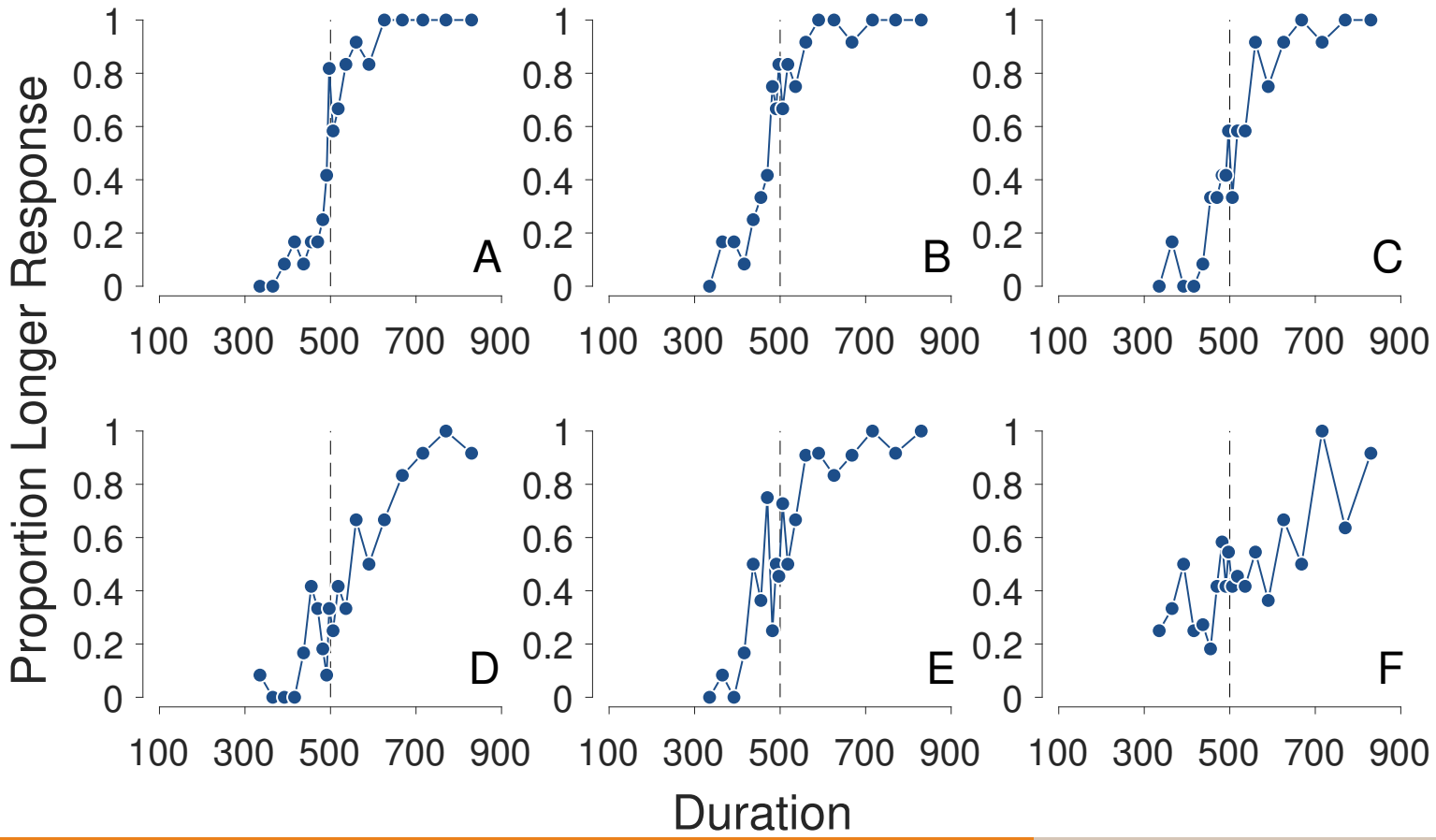
- Generalization involves model predictions about behavior for different gambles



Duration Perception

Duration Perception Experiment

- van Driel et al. (2014) studied duration perception
  - 240 trials in auditory and visual modalities
  - 20 stimuli varying from 335ms to 830ms each presented 12 times
  - always compared in duration to a 500ms standard
- The behavior of 6 participants in the visual task are shown

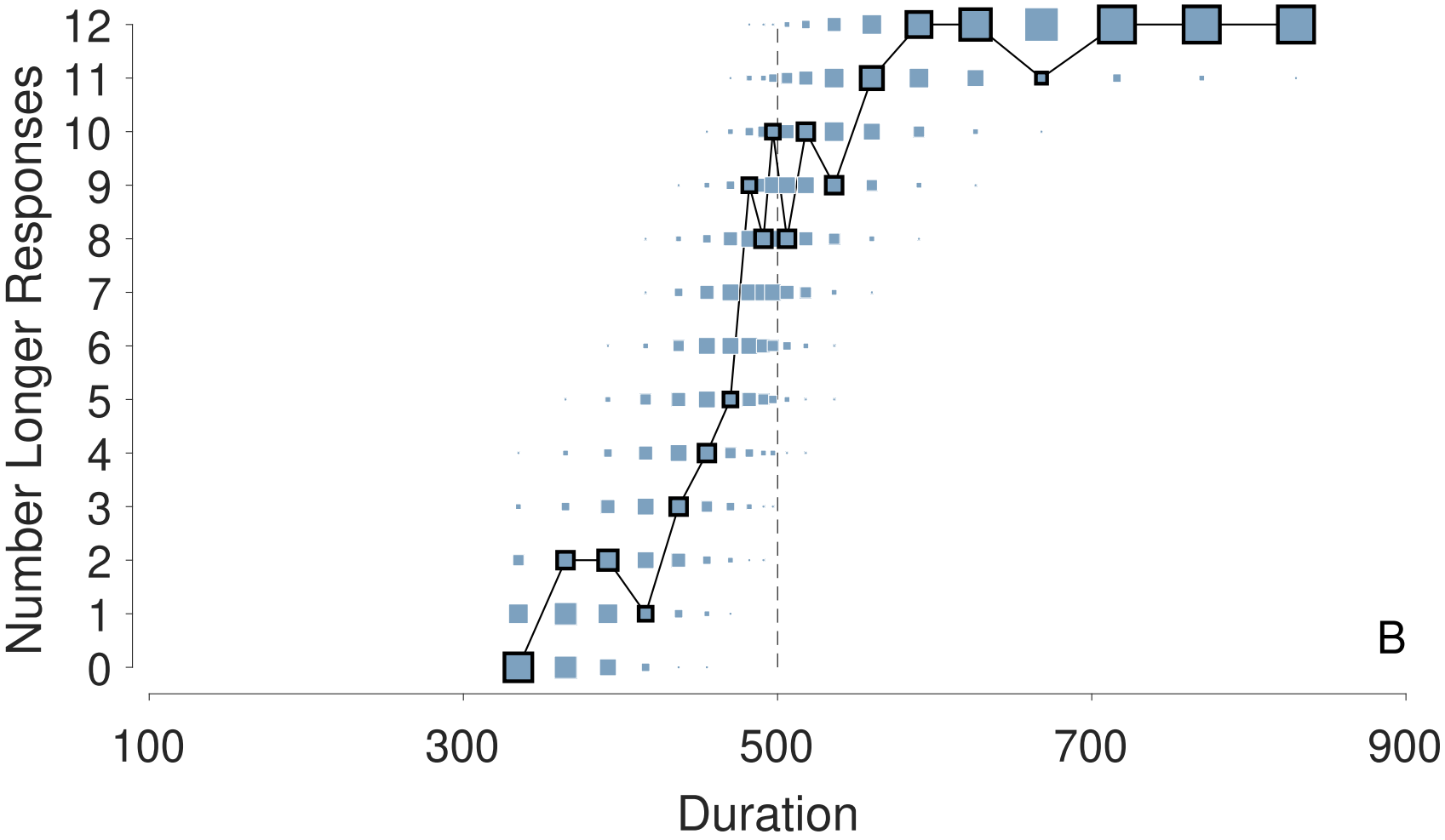


Applying the Logistic Choice Model

- The logistic choice model observed data are the A_t and B_t evidence values for each alternative on each trial
- In the Mixed Gambles example
  - A_t is the gain amount G, since larger values are more evidence for accepting the gamble
  - B_t is the gain amount L, since larger values are more evidence for not accepting the gamble
  - the interpretation of $A_t - \omega B_t$ is that ω is the weight of losses
- In this Duration Perception example
  - A_t is the duration of the stimulus on trial t, since larger values are more evidence for responding "longer"
  - B_t is the 500ms standard, since that duration is the is evidence against responding "longer"
- The interpretation of $A_t - \omega B_t$ is that ω is a response bias
  - is $\omega > 1$ the standard is given too much emphasis and there is a bias towards the "shorter" response
  - is $\omega < 1$ the standard is given too little emphasis and there is a bias towards the "longer" response

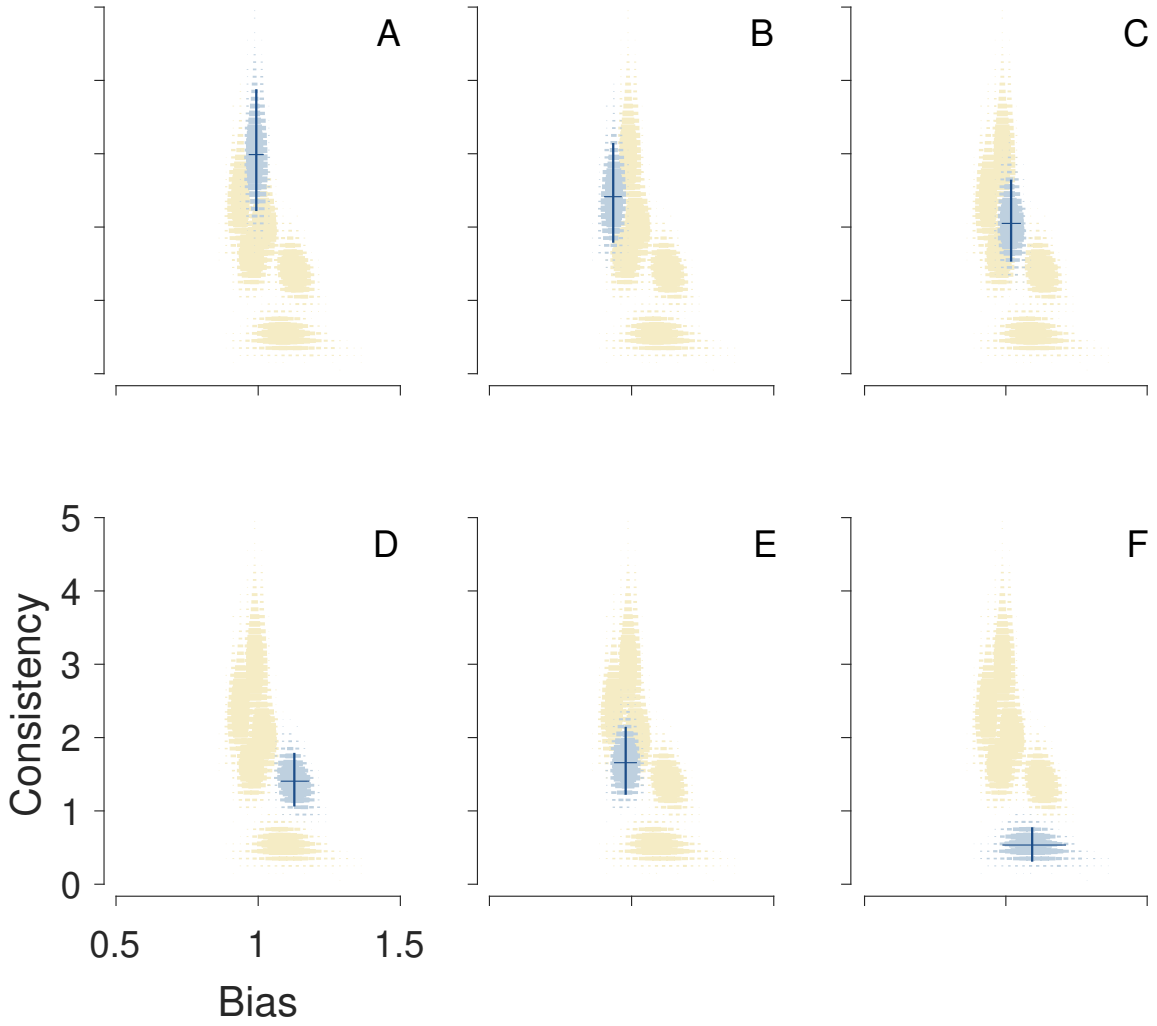
Descriptive Adequacy

- The blue squares show the posterior probability of the number of "longer" responses to each stimulu for participant B
  - the observed numbers are shown by the black borders



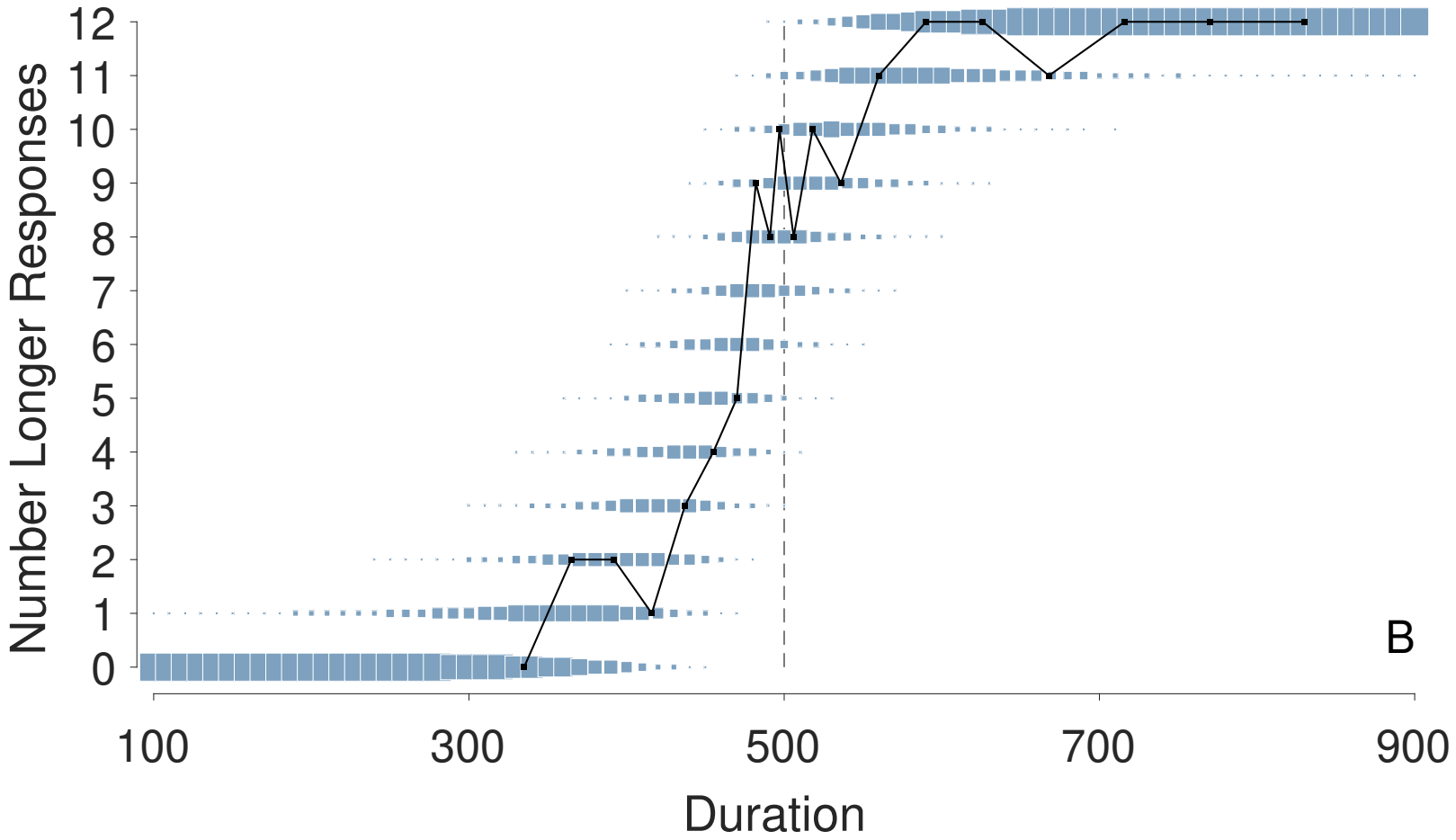
Consistency and Bias Inferences

- Each panel shows the joint posterior distribution for consistency λ and bias ω of a participant in blue
  - other participants are shown in yellow for comparison



Generalization

- A posterior predictive analysis that studies generalization uses durations other than the ones presented in the experiment
  - in the figure, all durations from 100ms to 900ms in increments of 10ms are considered



References i

Kahneman, D., & Tversky, A. (1984). Choices, values, and frames. *American psychologist*, *39*(4), 341. doi: 10.1017/cbo9780511803475.002

Pelli, D. G., & Farell, B. (1995). Psychophysical methods. *Handbook of optics*, *1*, 29–1.

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Wang, S., Yu, R., Tyszka, J. M., Zhen, S., Kovach, C., Sun, S., ... others (2017). The human amygdala parametrically encodes the intensity of specific facial emotions and their categorical ambiguity. *Nature communications*, *8*(1), 14821. doi: 10.1038/ncomms14821